

JEE MAIN REVISION MASTER: DIFFERENTIATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 83 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Continuity & Differentiability at a Point	52 PYQs	Continuity & Differentiability
Very High	Evaluation of Limits & L'Hopital's Rule	31 PYQs	Limits

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Continuity & Differentiability at a Point	$\text{Continuous at } a \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a), \quad \text{Differentiability} \implies \text{Continuity}$
Evaluation of Limits & L'Hopital's Rule	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1, \quad \lim_{x \rightarrow a} f(x)^{g(x)} = e^{\lim_{x \rightarrow a} (f(x) - 1)g(x)} \text{ (for } 1^\infty \text{)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Continuity & Differentiability at a Point (Very High Yield — 52 PYQs)

Core Principle: Epsilon-delta and limit criteria for continuity, left/right hand derivatives, and graphical identification of non-differentiable sharp points.

■■ **Caution / JEE Pitfall:** Differentiability implies continuity, but continuity DOES NOT imply differentiability (e.g. $f(x) = |x|$ is continuous at $x=0$ but not differentiable).

■ **Speed Trick:** The greatest integer function $[x]$ is discontinuous at all integer points $x \in \mathbb{Z}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "If $y = \tan^{-1}((3\cos x - 4\sin x) / (4\cos x + 3\sin x)) + 2\tan^{-1}(\frac{x}{1+\sqrt{1-x^2}})$, then (dy/dx) at $x = \frac{\sqrt{3}}{2}$ is e..." [Ans: N/A]
- [2025 Evening Shift] Q#1 (MCQ): "Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a function which is differentiable at all points of its domain and satisfies the condition $x^2 f'(x) = 1$..." [Ans: N/A]
- [2025 Morning Shift] Q#1 (Numerical): "Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0, f''(x) = f(x), f(0) = 0, f'(0) = 3$. Then $9f(\log_2 \dots)$ " [Ans: 36]

3.2 Evaluation of Limits & L'Hopital's Rule (Very High Yield — 31 PYQs)

Core Principle: Resolution of indeterminate forms ($0/0$, ∞/∞ , 1^∞), standard trigonometric and exponential limits, and Series expansion.

■ **Caution / JEE Pitfall:** Before applying L'Hopital's rule, ALWAYS check that the limit is in an indeterminate form ($0/0$ or ∞/∞). Differentiating a determinate form produces an incorrect answer.

■ **Speed Trick:** Maclaurin expansions are often faster than repeated L'Hopital: $\sin x = x - (x^3/6)$, $\cos x = 1 - (x^2/2)$, $e^x = 1 + x + (x^2/2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3)$, $x \in \mathbb{R}$. Then the value of $f'(5)$ is" [Ans: N/A]
- [2024 Evening Shift] Q#2 (MCQ): "If $\log_e y = 3 \sin^{-1} x$, then $(1-x^2)y'' - xy'$ at $x = (1/2)$ is equal to..." [Ans: N/A]
- [2024 Morning Shift] Q#4 (MCQ): "Let $f(x) = x^5 + 2e^{x/4}$ for all $x \in \mathbb{R}$. Consider a function $g(x)$ such that $(g \circ f)(x) = x$ for all $x \in \mathbb{R}$. Then the value of..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 2nd April Morning Shift (Q#1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0$, $f'''(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____.

Correct Answer: 36

Solution Analysis: $f'(x) \geq 0$, $f'''(x) = f(x)$ Second order differential equation $f(x) = Ae^x + Be^{-x}$ $f(0) = 0 \Rightarrow A = -B \Rightarrow f(x) = A(e^x - e^{-x})$ $f'(x) = Ae^x + Ae^{-x} = A(e^x + e^{-x})$ $f'(0) = 3 = A(e^0 + e^0) = 2A \Rightarrow A = (3/2)$ $f(x) = (3/2)(e^x - e^{-x})$ If $\ln 3 = (27/2)(e^{\ln 3} - e^{-\ln 3}) = (27/2)(3 - (1/3)) = (27/2) \cdot (8/3) = 36$

Benchmark Problem 2: JEE Main 2024 (Online) 27th January Morning Shift (Q#4)

Let $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in \mathbb{R}$. Then $f'(10)$ is equal to _____.

Correct Answer: 202

Solution Analysis: $f(x) = x^3 + x^2 \cdot f'(1) + x \cdot f''(2) + f'''(3)$ $f'(x) = 3x^2 + 2x f'(1) + f''(2)$ $f''(x) = 6x + 2f'(1)$ $f'''(x) = 6$ $f'(1) = -5$, $f''(2) = 2$, $f'''(3) = 6$ $f(x) = x^3 + x^2 \cdot (-5) + x \cdot (2) + 6$ $f'(x) = 3x^2 - 10x + 2$ $f'(10) = 300 - 100 + 2 = 202$